

WASP-52b

R-1686

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-52_150924 · created 2026-08-02T12:30:03+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2457289.7465 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.214 +/- 0.013
Transit depth (Rp/Rs)^2	4.59 +/- 0.56 [%]
Orbital Inclination (inc)	83.53 +/- 0.44
Airmass coefficient 1 (a1)	1.148 +/- 0.01
Airmass coefficient 2 (a2)	-0.12 +/- 0.017
Scatter in the residuals of the lightcurve fit is	0.88 %
Best Comparison Star	None
Optimal Aperture	3.94
Optimal Annulus	11.88
Transit Duration (day)	0.0671 +/- 0.0042

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.214 $\pm$ 0.013 vs 0.1646 (deviation 3.8 σ)
Duration fit vs published	0.0671 d vs 0.0758 d (deviation 2.08 σ)
Mid-time O-C	2.6 min
Depth SNR	16.5
Residual scatter	0.88 %

Light curve

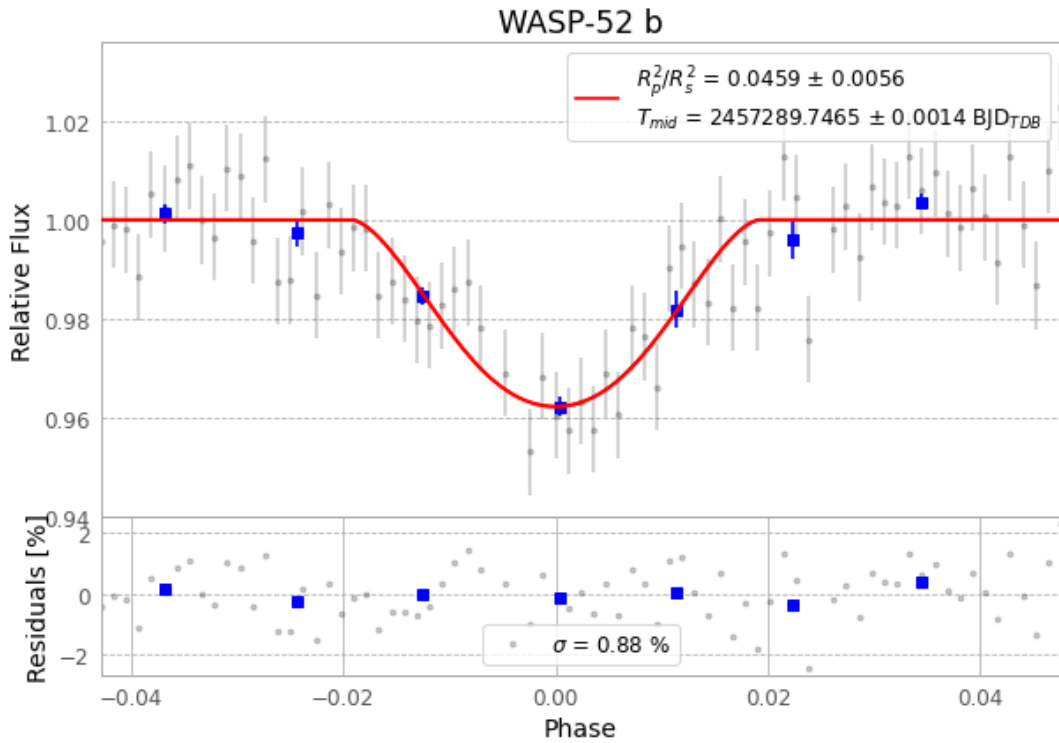


Figure 1 — FinalLightCurve_WASP-52 b_2015-09-23.png (SHA-256 03fd9f1c975e249d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [419, 255], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 320d9a56e68a6a18...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

77 FITS frames (51 MB), WASP-52150924035712.FITS → WASP-52150924074512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-150924.log (54c292bee958...), FinalLightCurve_WASP-52 b_2015-09-23.png (03fd9f1c975e...), FinalLightCurve_WASP-52 b_2015-09-23.csv (812ff25f7cea...)

Compute resources

Wall time 145.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 12:30Z.